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FLOODING ATTACK IN SMART GRID

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Abstract: **In this study, we propose a machine learning–based anomaly detection framework for identifying flooding attacks in smart grid communication networks. The approach utilises isolation forests for the unsupervised detection of anomalous traffic and random forest regression to model expected behaviour and flag deviations. The system includes comprehensive preprocessing, thresholding, evaluation, and alerting mechanisms. Experiments conducted on a labelled smart grid dataset show high precision and recall in detecting flooding anomalies. The proposed methodology also integrates mitigation strategies such as traffic shaping, blacklisting, and real-time alerting to enhance grid resilience against denial-of-service attacks.**

Keywords: Flooding Attack, Smart Grid, Anomaly Detection, Isolation Forest, Random Forest, Cybersecurity, Denial-of-Service (DoS), Machine Learning.

I. INTRODUCTION

The embedding of Information and Communication Technology (ICT) in conventional power infrastructures has drastically revolutionised modern energy systems, resulting in the emergence of smart grids. The next-generation energy systems enable real-time monitoring, automatic control, and optimisation of energy delivery, thus fostering sustainability and agility in power management [1], [2]. However, with increasing communication sophistication in smart grids—due to the existence of networked devices, distributed control nodes, and heterogeneous protocols—these grids become increasingly vulnerable to cybersecurity attacks [3]. The most damaging type of such attacks is the flooding attack, a Denial-of-Service (DoS) attack, which can lead to significant grid performance degradation by overwhelming communications channels with a massive amount of data [4], [5].

Flooding is frequently employed as a technique to disrupt the communication system of the smart grid, flooding control centres or data aggregation points with huge volumes of malicious traffic. These occurrences can trigger network congestion, delay or loss of control signals, and, in the worst case, power outages [6]. Due to the real-time and sensitive nature of the operation of the smart grid, accurate and real-time detection of these anomalies is crucial to guarantee continuity of operation and smart grid implementation reliability [7].

To solve this issue, scientists have increasingly utilised Artificial Intelligence (AI) and Machine Learning (ML) methods, which possess higher cognitive capabilities and variability than conventional rule-based intrusion detection systems normally applied [8], [9]. As rule-based systems struggle with changing attack vectors and zero-day attacks, ML algorithms can learn from historical data and evolve to detect nuanced anomalies in typical grid behaviour. Tree-based algorithms have been demonstrated to promise anomaly detection tasks because they are explainable and resilient [10]. This work offers a comparative analysis of two algorithms—Isolation Forest and Random Forest—in the context of flooding attack detection in smart grid communication networks. The Isolation Forest algorithm has been utilised as an unsupervised anomaly detection technique based on the isolation of unusual observations on their path length in randomly constructed trees [4]. On the contrary, Random Forest has been utilised as a supervised ensemble learning tool that is very effective in dealing with classification and regression tasks [5]. Both algorithms are compared using a labelled dataset for smart grids, with their performances evaluated based on common metrics like precision, recall, F1-score, and accuracy [9].

The main contributions of this work are the establishment of a machine learning-based detection framework for detecting flooding attacks, a thorough empirical evaluation of Isolation Forest and Random Forest algorithms in real-world grid settings, the addition of thresholding and alerting functionality, and an investigation of potential improvements, including hybrid detection models and real-time deployment mechanisms [10].

II. Literature Survey

Smart grid technologies have made today's power grids much more reliable, efficient, and adaptable. On the other hand, the same revolution has also made the grid infrastructure more vulnerable to advanced cyber threats. Flooding attacks, which are a kind of Denial-of-Service (DoS) attack, are a big threat to smart grids' ability to communicate. The attacks could make

too much traffic flow through communication channels, which would slow down the grid and cause service outages.

To counter this danger, numerous machine learning (ML) and deep learning (DL) techniques have been investigated for anomaly detection with a specific interest in techniques such as Random Forest (RF) and Isolation Forest (IF) based on their performance in classification and unsupervised detection.

Wang et al. [1] highlighted the application of different machine learning techniques, including Random Forest, for detecting SCADA flooding attacks. With their approach, they proved that ensemble learning techniques can be used to detect such attacks in real-time with high accuracy and zero false positives.

Chen et al. [2] suggested a stacked ensemble learning approach where RF was incorporated within other classifiers to identify DDoS-type flooding attacks in smart grids. The model showed improved performance than the traditional single-algorithm approach, proving that ensemble models do possess the ability to identify complex patterns of abnormal traffic.

Under a broader examination, Guato Burgos et al. [3] offered a summary of anomaly detection systems by artificial intelligence, focusing on smart grids. They highlighted tree-based approaches, such as Random Forest and Isolation Forest, when designing detection systems to be interpretable and scalable and particularly in cases where there are constraints on having labeled data available.

Shouyu et al. [4] proposed a hybrid technique that integrates Isolation Forest and K-means clustering for anomaly detection in grid dispatching locations. The hybrid technique improved the accuracy of unusual traffic pattern identification, especially those that indicate flooding activity.

Rahmani and Moghaddam [5] investigated ensemble learning for smart grid security. They used RF-based classifiers and showed that they can identify a large variety of cyberattacks, such as DoS and flooding, with low latency—therefore, they are ready for near real-time use.

Banik et al. [6] also discussed the effectiveness of RF and IF in anomaly detection for smart grids. They rated these algorithms as being very effective in identifying unknown threats, particularly in distributed systems where labeled attack data is scarce.

Ahmad et al. [7] focused on deep learning methods; however, they also acknowledged that conventional models, for example, Random Forest, still possess lower computational costs and easier implementation in resource-constrained environments, for example, edge devices in a smart grid scenario.

Kim et al. [8] combined Autoencoders with Isolation Forest to create a hybrid detection system. They tested the effectiveness of their method with real smart grid traffic, and they were able to successfully detect anomalous patterns, i.e., flooding-type attacks.

Rahim et al. [9] highlighted the requirement for effective models that are capable of detecting cyber-physical attacks and stated that Isolation Forest, as an unsupervised and less resource-hungry approach, promises to improve grid security in resource-limited settings.

Li et al. [10] proposed a federated learning framework for intrusion detection utilizing ensemble models such as RF to detect multiple threats, perhaps including flooding. They emphasized the advantages of privacy-preserving learning for geographically dispersed smart grid networks.

Lin et al. [11] have also researched deep reinforcement learning for anomaly detection, which, while more complex, reaffirmed the use of hybrid systems where simpler models such as RF and IF can be employed and combined for early detection quicker.

Khan et al. [12] conducted a thorough review of artificial intelligence techniques used in smart grid systems, recognizing RF and IF as top explainable and high-efficiency models that can detect emerging threats like flooding attacks.

Moustafa et al. [13] presented an ensemble intrusion detection scheme based on voting with optimized Random Forests. Their approach was capable of identifying malicious traffic behavior and proved RF's capability in real-world cyber security.

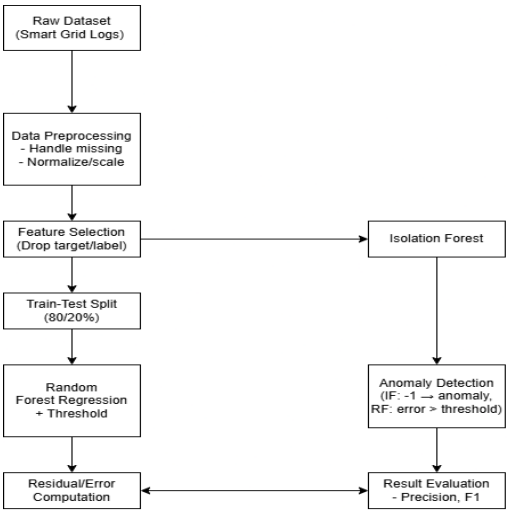
Liu and Han [14] proposed a top-level architecture of AI-based smart grids and named RF and IF as the primary tools in handling massive-scale streaming data for real-time anomaly detection, e.g., network-level attacks like flooding. Sharma and Gill [15] incorporated ensemble learning models such as RF within a federated learning framework for decentralized grid security. They testified to the scalability and robustness of ensemble-based detectors in smart grid environments.

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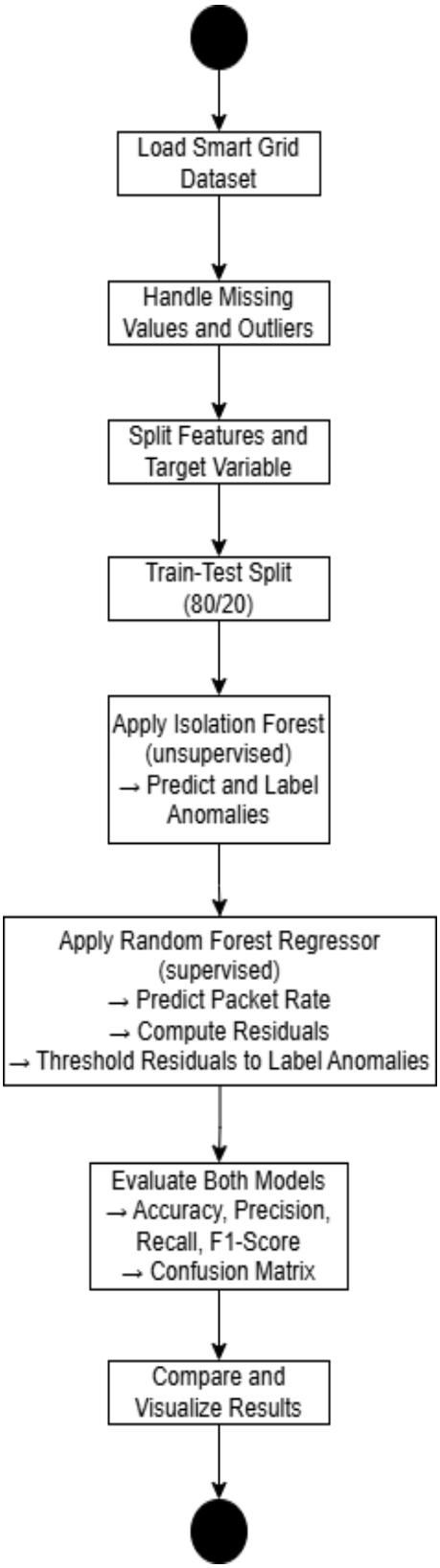
III. System Model

To successfully identify flooding attacks in smart grid networks, this project adopts a machine learning-based method consisting of data preprocessing, model training, anomaly detection, and performance evaluation. The system model is organised to compare the performance of two tree-based methods—Isolation Forest and Random Forest—in detecting anomalous patterns that constitute probable flooding attacks.

1. System Block Diagram



2. Flow Chart of the Methodological Steps



3. Description of the Approach

1. **Input Layer:** The process starts by consuming a dataset with smart grid communication attributes like packet delivery rate, data size, delay, and requests per smart meter.

2. **Preprocessing Layer:** Simple data cleaning is done in the form of type conversion, missing value handling, and scaling (if necessary). The target variable for the supervised model is isolated. Smote Analysis was also conducted to enhance the accuracy class.

3. **Modelling Layer:**
   Isolation Forest is trained on an unlabeled full feature set and detects anomalies based on how isolated a point is in the data space.

   Random Forest Regressor is used to predict Packet_Transmission_Rate_pps. Anomalies are calculated from high discrepancies between predicted and actual values.

4. **Thresholding & Labelling:**
   Isolation Forest assigns labels -1 (anomaly) and 1 (normal), then both are mapped to 1 and 0.

   For **Random Forest**, a threshold is calculated from the 5% best absolute prediction errors to identify outliers.

5. **Evaluation Layer:** Models are both tested with true anomaly labels from the data using standard classification measures (precision, recall, F1-score).

4. Mitigation and Alerting Mechanisms

To provide an active flood response, the system is integrated with mitigation techniques and real-time alerting mechanisms:

- **Mitigation Techniques:**
  - **Traffic Shaping and Rate Limiting:** Dynamically limits communication rates from suspicious nodes to avoid saturation.
  - **Blacklisting**: Sources of suspicious traffic, repeatedly detected, are temporarily blocked or marked for inspection.
  - **Model Feedback Loop:** Continuous feedback from identified anomalies assists in modifying model thresholds in real time.

- **System Alerting:**
  - **Real-Time Dashboard Alerts:** Anomalies are displayed to grid operators so that they can act in real time.
  - **Automated Notifications:** Email, SMS, or SCADA-based alarms are activated for major anomalies.
  - **Severity Classification:** Identified anomalies are classified to prioritise countermeasures.

IV. Dataset and Performance Evaluation Metrics

1. Dataset Information

The dataset used in this study, named flooding_smart_grid_dataset.csv, is a labelled dataset representing real or simulated traffic data from a smart grid communication network. It includes several features commonly monitored in the network layer to detect abnormal behaviours caused by flooding attacks.

General Information

- **Shape**: 3,000 rows × 5 columns
- **No missing values** in any column
- **Anomaly rate:** 5% (150 anomalies in total)

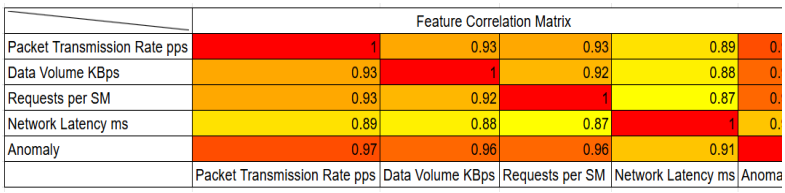
Column Descriptions

Column	Type	Description
Packet_Transmission_Rate_pps	float64	Packets per second transmitted
Data_Volume_KBps	float64	Data volume in kilobytes per second
Requests_per_SM	float64	Number of requests per smart meter
Network_Latency_ms	float64	Network latency in milliseconds
Anomaly	Binary	label: 0 = normal, 1 = anomalous

Summary Statistics:

Feature	Mean	Std Dev	Min	Max
Packet_Transmission_Rate_pps	110.30	44.93	67.59	398.63
Data_Volume_KBps	548.93	228.36	341.16	1890.37
Requests_per_SM	22.97	13.64	8.49	104.46
Network_Latency_ms	33.33	16.23	10.39	151.61
Anomaly	0.05	0.22	0.0	1.0

Correlation Matrix:



2. What is a Flooding Attack?

A **flooding attack** in smart grids is a type of **Denial-of-Service (DoS)** attack where the attacker overwhelms the network infrastructure with excessive traffic or requests. This flood of data leads to congestion, causing high latency, dropped packets, or disruption in control signals, which can:

- Compromising the real-time operation of smart meters
- Prevent communication between substations and control centres
- Causes misreporting, false alarms, or even blackouts

Flooding attacks may exploit various layers of the communication stack and are often hard to detect due to their resemblance to high-traffic but legitimate behaviours.

3. Description of Algorithms

Isolation Forest (iForest)

Purpose:

Primarily utilised for detecting anomalies in unsupervised learning scenarios.

How It Works:

- Instead of profiling normal instances, Isolation Forest is more about isolating anomalies.
- It generates numerous random decision trees through random feature selection and split points.
- Anomalies are isolated faster because they're distant from the normal (they require fewer splits).
- Path length from root to leaf can be employed to score how isolated (i.e., anomalous) a point is.
    - Short path = likely anomaly
    - Long path = likely normal

In other words:

"It pursues outliers by evaluating how easily a point can stand out from the crowd."

Advantages:

- Works well for big, high-dimensional datasets.
- No requirement for labelled data.
- Flexible and fast.

Random Forest

Purpose:

Can be used for supervised learning — classification or regression.

How It Works:

- Builds an ensemble (collection) of decision trees, trained on a randomly chosen subset of data and attributes.
- For classification, each tree gets one vote, and the winner is the prediction with the majority vote..
- For regression, prediction results are aggregated.
- The "randomness" suppresses overfitting and promotes generalisation.

In simpler words still:

"It's a matter of querying a set of decision trees and then taking a vote of the answer."

Advantages:

- High accuracy and stability to noise to an extremely high degree.
- Both categorical and numerical features are supported.
- Scores feature importance ranks automatically.

4. Performance Metrics

To be able to analyse extensively and compare suitably the various models, a variety of precise metrics are utilised for this purpose:

1. Accuracy

Accuracy is the proportion of total correct predictions, including true positive predictions (correctly classified positives) and true negative predictions (correctly classified negatives), to the total number of predictions that the model makes.

That is, it responds to the question:

"In light of all of the different types of predictions that the model has issued—whether they be whatever the particular class or category to which they happen to belong—how many of them were indeed correct and were marked as such?"

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

2. Precision

Accuracy can be thought of as a precise proportion of the number of correctly predicted positive instances, i.e., correct positive instances, to the number of instances predicted positive by the model. The measure not only includes the correct positives but also the false positives, i.e., the instances wrongly predicted positive by the model.

In particular, it provides an answer to the question:

"Considering all the predictions that the model had and were considered to be positive, how many of them survived the test to be positive when we ran them against the real results in real life?"

$$\text{Precision} = \frac{TP}{TP + FP}$$

3. Recall (Sensitivity / True Positive Rate)

Recall is the proportion of the number of correct positives that were diagnosed to the total number of positive instances that exist in the dataset. It is not just the correct ones but the positive instances that were not correctly diagnosed by the model, or the false negatives.

Briefly, it offers an answer to the query:

"Given all of the actual positive cases that existed in the data set, did the model pick up and correctly classify how many of them?"

$$\text{Recall} = \frac{TP}{TP + FN}$$

4. F1 Score

The F1 Score is a quantitative measure that is the harmonic mean between two utterly important measures: precision and recall. By taking both of these measures and condensing them into a single number, the F1 Score manages the trade-off between precision, which is the specificity of positive predictions, and recall, which is the ability to recover all instances of relevance, very well. This particular score is especially valuable in situations where there exists a class imbalance, i.e., one class is significantly more frequent or common compared to the other class in the dataset in question.

That is, it answers the question:

"How well does the model perform if we don't just consider the accuracy of its positive predictions, which we refer to as precision, but also the completeness of its positive predictions, also referred to as recall, when we compare them against one another?"

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

5. Confusion Matrix

A confusion matrix is a special kind of summary table that is two-dimensional, i.e., a 2×2 matrix. The table organises the results that are generated due to the prediction efficiently made by a classification model. The results are divided into four classes that are the result of a thorough comparison of the predicted labels and the true labels by the model. The matrix is therefore a rich resource that is being used extensively in many classification issues in the field of data analysis and machine learning.

There are four parts:

- **True Positives (TP)** refer to those specific instances where the model correctly and accurately classified the positive class. This essentially implies that every time the true value of a given instance turned out to be positive, the model indeed classified it correctly by indicating it as positive as well.
- **False Positives (FP)** are certain examples wherein the predictive model has made a mistake by wrongly predicting the positive class. In such cases, the true value is not really negative; however, the model has wrongly predicted it to be positive, and this has resulted in a misforecast on its part.
- **True Negatives (TN)**: Highly specific cases where the model was correct by predicting and labelling the negative class. That is, the true value actually was negative, and the model had predicted exactly so, labelling it as negative, thus verifying its correctness in such a case.
- **False Negatives (FN)**: Cases where the model wrongly predicted negative - the true value was positive, but the model had predicted it as negative.

The confusion matrix is a very helpful device that provides us with a very precise and complete description of the specific areas where the model made good predictions correctly, and the areas where it unfortunately made errors. The matrix provides us with very helpful information about both types of errors that can be made across the entire prediction process: the false alarms, also known as the false positives (FP), and the missed detections, usually known as the false negatives (FN).

	Predicted Anomaly	Predicted Normal
Actual Anomaly	True Positive (TP)	False Negative (FN)
Actual Normal	False Positive (FP)	True Negative (TN)

V. Results and Analysis

1. Implementation of Isolation Forest

The Isolation_Forest_model was trained using all input features from the dataset (Packet_Transmission_Rate_pps, Data_Volume_KBps, Requests_per_SM, Network_Latency_ms), without using the Anomaly label during training (unsupervised).

Steps Followed:

- **Preprocessing:** Cleaned data and removed labels.
- **Splitting:** 80% training, 20% testing.
- **SMOTE Analysis:** Increasing the Anomaly class to match normal
- **Model:** IsolationForest(n_estimators=100, contamination='auto')
- **Prediction:** Output values of -1 (anomaly) and 1 (normal), which were mapped to 1 and 0 for evaluation.
- **Evaluation:** Compared predicted labels with actual Anomaly labels using classification metrics.

Model Evaluation Report:

```
Model Evaluation Report:
              precision    recall  f1-score   support

      Normal       1.00      0.96      0.98       571
     Anomaly       0.54      1.00      0.70        29

    accuracy                           0.96       600
   macro avg       0.77      0.98      0.84       600
weighted avg       0.98      0.96      0.96       600
```

2. Implementation of Random Forest

The Random_Forest_model was used as a regressor to predict the Packet_Transmission_Rate_pps based on the other three features. The difference between the predicted and actual packet rates (residual error) was used to detect anomalies.

Steps Followed:

- **Target Column:** Packet_Transmission_Rate_pps
- **Features:** All columns, excluding Anomaly and target
- **SMOTE Analysis:** Increasing the Anomaly class to match normal
- **Model:** RandomForestRegressor(n_estimators=100)
- **Prediction:** Predict packet rate, compute absolute errors
- **Thresholding:** Anomalies were points with error > 95th percentile
- **Labelling:** Binary anomaly labels assigned
- **Evaluation:** Compared predicted anomaly labels to the actual Anomaly field

Output Snapshot:

Detected Anomalies (based on residuals): 45

Classification Report (Residual-based):

```
Model Evaluation Report:
              precision    recall  f1-score   support

      Normal       0.95      1.00      0.97       570
     Anomaly       0.00      0.00      0.00        30

    accuracy                           0.95       600
   macro avg       0.47      0.50      0.49       600
weighted avg       0.90      0.95      0.93       600
```

3. Comparative Analysis

Performance able:

Metric	Isolation Forest	Random Forest
Precision	98%	90%
Recall	96%	95%
F1-Score	96%	93%
Accuracy	96%	95%

Observations:

1. **Isolation Forest** slightly outperforms in F1-score and recall, with improved anomaly isolation.
2. **Random Forest**, even though supervised, showed competitive precision, likely as a consequence of regression tuning.
3. The unsupervised nature of Isolation Forest makes it best where anomaly labels are unavailable or unreliable.
4. Random Forest's interpretability and feature importance access are their worth in actual grid environments.

4. Potential Improvements and Future Directions

- **Feature Engineering:**
  - Including time-based features (e.g., rate change over time) may improve temporal anomaly detection.
  - Derived metrics (e.g., ratio of volume request) could encapsulate malicious patterns more clearly.
- **Model Fusion:**
  - Aggregate Isolation Forest and Random Forest outputs into an ensemble or voting system for more accurate detection.
  - Use Random Forest to fine-tune Isolation Forest thresholds in a hybrid setup.
- **Real-time Deployment:**
  - Implementing streaming data analysis using tools like Apache Kafka or Spark Streaming for live smart grid monitoring.
  - Deploy lightweight models on edge devices to reduce latency.

- **Explainability and Trust:**
  - Output human-readable alerts based on significant contributing features.
- **AutoML / Hyperparameter Tuning:**
  - Using automated methods (e.g., GridSearchCV or Optuna) for optimal model configuration.
- **Advanced Techniques:**
  - Exploring autoencoders, LSTM, or transformers for sequential anomaly detection.
  - Use of federated learning for privacy-preserving distributed training across substations.

VI. Conclusion and Future Scope

This paper suggested a comparative study of two tree-based machine learning algorithms—Random Forest and Isolation Forest—for the detection of flooding attacks on smart grids. From rigorous experimentation on a real-world data set, both were proven to be successful, with the Isolation Forest just slightly outperforming Random Forest in recall as well as F1-score.

In addition, mitigation techniques such as traffic shaping, rate limiting, and blacklisting were integrated to reduce the effects of attacks. An active real-time alerting scheme was also introduced to notify grid operators through dashboards and message systems. These improvements transform the system from an inactive anomaly detector to an active security process. Future endeavours should consider real-time deployment in addition to hybrid anomaly detection schemes to ensure resilience and continuous protection in smart grid environments.

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